

CAPSTONE PROJECT REPORT

Report 5 – Software Test Documentation

– Hanoi, Nov 2025 –

Table of Contents

I. Record of Changes	3
II. Testing Documentation.....	4
1. Scope of Testing	4
1.1 Target of Test (features & functions).....	4
1.2 Non-functional requirements	4
2. Test Strategy	5
2.1 Testing Types.....	5
2.2 Test Levels	5
2.3 Supporting Tools	5
3. Test Plan.....	5
3.1 Test Environment.....	5
3.2 Test Milestones	6
4. Test Cases.....	6
4.1 Unit Test.....	6
4.2 Integration Test.....	6
4.3 System Test	6
5. Test Reports	7
5.1 Unit test report	7
5.2 Integration test report	7
5.3 System test report.....	8

I. Record of Changes

Date	A* M, D	In charge	Change Description
5/12/2025	A	QuatBT	Add Test Documentation
12/11/2025	A,M	TuanCM	Add contents, reports
14/11/2025	M	Team G44	Update Test Documentation
19/12/2025	M	Team G44	Update Test Documentation
10/12/2025	M	Team G44	Update Test Documentation
12/12/2025	M	Team G44	Update Test Documentation

*A - Added M - Modified D - Deleted

II. Testing Documentation

1. Scope of Testing

1.1 Target of Test (features & functions)

The tests will cover the OLP-SkillUp features below (primary targets):

- **Course Management:** Course creation, editing, publishing (draft/published states), curriculum organization with sections and lessons, course categories and subcategories, course enrollment and progress tracking.
- **Assessment & Quizzes:** Quiz creation with multiple question types, timer and passing percentage configuration, quiz attempts and scoring, student progress tracking.
- **Student Progress & Enrollment:** Course enrollment, lesson completion tracking, progress percentage calculation, learning history and recently accessed courses.
- **Rating & Reviews:** Course ratings (star-based), written reviews and feedback, rating aggregation and average calculation, student-only rating permissions.
- **Lecturer Application System:** CV submission for becoming a lecturer, application status tracking (Pending/Approved/Rejected), admin review and approval workflow, lecturer profile with bio and expertise.
- **Payments:** Shopping cart functionality, course pricing (free/paid), voucher/coupon system with discount codes, transaction processing, payment method integration, order history.
- **User Management:** Registration, login, profile management, role-based access (Student, Lecturer, Content Moderator), password reset with OTP, token-based authentication (JWT + Refresh Token).

1.2 Non-functional requirements

The following non-functional areas will be validated at a basic/operational level:

- **Compatibility:** RESTful API architecture for cross-platform support, CORS configuration for frontend-backend communication, support for multiple browsers (modern browsers via Vite/React), mobile-responsive design with Tailwind CSS.
- **Security (basic):** JWT-based authentication with token expiration, role-based authorization (Student, Lecturer, Content Moderator, Admin), refresh token mechanism for session management, OTP verification for password reset, input validation and sanitization, secure file upload restrictions.
- **Reliability:** correct error handling and stable recovery for common failure cases.
- **Documentation:** correctness and completeness of user guides and API docs.

2. Test Strategy

2.1 Testing Types

2.2 Test Levels

Type of Tests	Test Level		
	Unit	Integration	System
Function Testing	X		X
Integration Testing		X	X
Performance Testing		X	X
API Testing	X	X	

2.3 Supporting Tools

Purpose	Tool	Vendor/In-house	Version
Test Result	Google sheets(excel)	Google	Latest
Front-end test	Chrome Developer/ Microsoft Edge	Google/Microsoft	Latest
Manage Bugs	Google sheets(excel)	Google	Latest

3. Test Plan

3.1 Test Environment

Purpose	Tool	Provider	Version
Test API	Postman	Postman.Inc	Lastest
Test UT	NUnit	.NET Foundation	3.14.0
Test IT	Chrome Developer/ Microsoft Edge	Google/Microsoft	Lastest

Test ST	Chrome Developer/ Microsoft Edge	Google/Microsoft	Latest
---------	----------------------------------	------------------	--------

3.2 Test Milestones

Milestone Task	Start Date	End Date
Test Planning	21/9/2025	12/12/2025
Unit Testing Execution	3/11/2025	12/12/2025
Integration Testing Execution	17/11/2025	12/12/2025
System Testing Execution	7/12/2025	12/12/2025
API Testing Execution	2/10/2025	12/12/2025
Test Report Preparation	11/12/2025	12/12/2025
Bug Fixing and Retesting	6/11/2025	12/12/2025

4. Test Cases

4.1 Unit Test

The unit testing phase focuses on verifying the core methods mentioned above.

For more full detail test cases: [Report5.1 Unit Test.xlsx](#)

4.2 Integration Test

Integration testing is performed by combining individual modules and examining them as a complete group. This phase aims to ensure that the interconnected components interact properly and that the system as a whole operates according to expectations.

For more full detail test cases: [Report5.2 Integration Test.xlsx](#)

4.3 System Test

- This testing approach focuses on observing and assessing how a complete system operates. It integrates various management processes to achieve the outcomes users need.
- Test cases serve as blueprints for system testing. They specify the actions performed within the system, the data provided as input, and the anticipated results at each stage of the test.

For more full detail test cases: [Report5.3 System Test.xlsx](#)

5. Test Reports

5.1 Unit test report

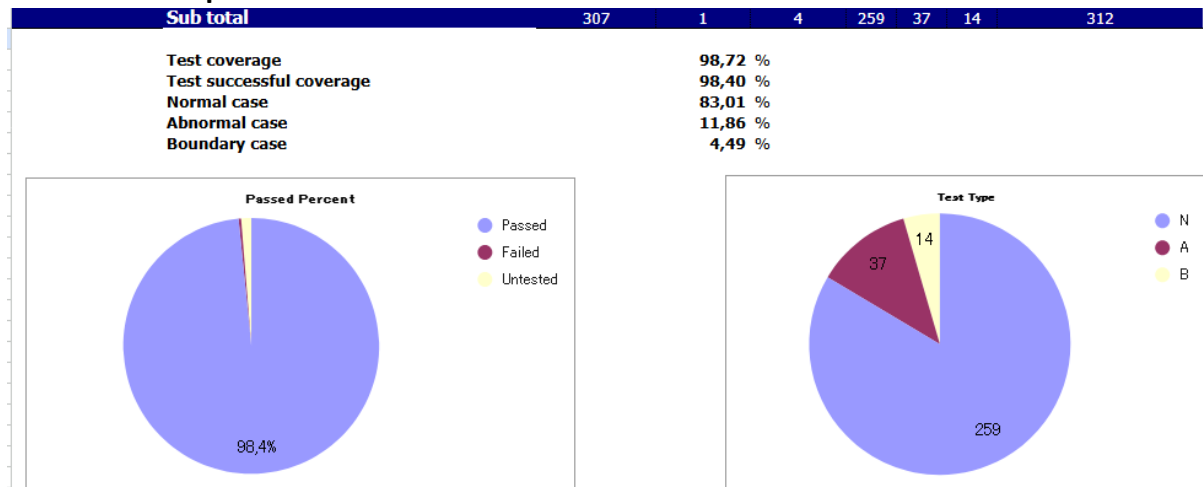


Figure 5.1: Unit test report

5.2 Integration test report

No	Module code	Passed	Failed	Pending	N/A	Number of test cases
1	Register	4	0	0	0	4
2	Login by account	3	0	0	0	3
3	Login by google	1	0	0	0	1
4	Forgot password	2	0	0	0	2
5	Logout	2	1	0	0	3
6	View profile	1	0	0	0	1
7	Update profile	2	0	0	0	2
8	Change password	4	0	0	0	4
9	View purchase history	2	0	0	0	2
10	View cart	2	0	0	0	2
11	Add to cart	3	0	0	0	3
12	Remove from cart	2	0	0	0	2
13	Add course	2	0	0	0	2
14	Hide course	1	0	0	0	1
15	Update course - general	9	0	0	0	9
16	View section	2	0	0	0	2
17	Add section	4	0	0	0	4
18	Edit section	3	0	0	0	3
19	Delete section	1	0	0	0	1
20	Create lesson	6	0	0	0	6
21	Create question	4	0	0	0	4
22	Add from bank	2	0	0	0	2
23	View question	2	0	0	0	2
24	Edit question	3	0	0	0	3
25	Delete question	2	0	0	0	2
26	Enroll course	3	0	0	0	3
27	Learning course	3	0	0	0	3
28	Progress tracking	1	0	0	0	1
29	Checkout	5	0	0	0	5
30	Apply voucher	5	0	0	0	5
31	Take quiz	5	0	0	0	6
32	View result	6	0	0	0	6
33	Rating	2	0	0	0	2
34	Email notification	4	0	0	0	4
Sub total		103	1	0	0	105

Test coverage **99,05 %**
 Test successful coverage **98,10 %**

Figure 5.2: Integration test report

5.3 System test report

TEST STATISTICS						
Project Name	Online Learning Platform – SkillUp	Creator		QuatBT		
Project Code	OLP-SkillUp	Reviewer/Approver				
Document Code	OLP-SkillUp_Test Report	Issue Date		2/11/2025		
Notes						

No	Module code	Passed	Failed	Pending	N/A	Number of test cases
1	Authentication	14	1	0	0	19
2	Workflow of System Moderator	15	0	0	0	18
3	Workflow of Content Moderator	18	0	0	0	20
4	Workflow of Lecturer	63	0	0	0	71
5	Workflow of Student	33	0	0	0	36
6	Workflow of Admin	16	0	0	0	17
7	AI assistant	5	0	0	0	6
Sub total		159	1	0	0	181

Test coverage 88,40 %
Test successful coverage 87,85 %

Figure 5.3: System test report